

Performance Testing

Date	01 july 2026
Team ID	SWTID-2026-2152
Project Name	Emotion Detection & Learning Support Engine
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Apache JMeter
Type of Testing	Load and Stress Testing
Target Module / API	Emotion Detection & Learning Engine API
Test Environment	Local Development Machine (Windows 11, 16GB RAM)
Test Date	30 jun 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Low Load: Single user processing	1	30	Stable performance
2	Moderate Load: Concurrent requests	20	60	No latency spike
3	Peak Load: Max expected capacity	50	120	System remains responsive
4	Stress Test: High saturation	100	180	Minimal error rate

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass/Fail)	Remarks
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1	Response Time (Avg)	< 2 seconds	0.8s	Pass	Excellent
2	Response Time (Max)	< 5 seconds	2.1s	Pass	Stable
3	Throughput	-	45 req/sec	Pass	Optimal
4	Error Rate	< 1%	0.05%	Pass	Negligible
5	CPU Utilization	< 80%	62%	Pass	Efficient
6	Memory Utilization	< 80%	55%	Pass	Stable

Step 4: Observations & Analysis

Key Findings

The performance testing results demonstrate that the system successfully meets all defined operational benchmarks under the current load. Specifically, the system exhibits stable response times (0.8s average, 2.1s maximum) and efficient resource utilization, with CPU usage at 62% and memory usage at 55%, both well within the 80% threshold. Furthermore, the application maintains high reliability, evidenced by an optimal throughput of 45 req/sec and a negligible error rate of 0.05%.

Bottlenecks Identified

No performance bottlenecks were identified during the testing process. All tested metrics consistently remained within the acceptable target values, indicating that the system's architecture is currently well-balanced for the tested workload.

Optimization Steps Taken

As all system performance metrics successfully passed the defined criteria, no specific corrective optimizations were required. The system configuration is currently optimized to handle the established traffic and resource demands.

Step 5: Screenshots / Evidence



Emotion-Aware Learning Assistant Framework

 Input Form Validation Interface

 Live Operational Inference Metrics

 Cumulative Historical Analytics View

Interactive Session Query Interface Node

What Field are you studying?

Computer Science

Target Active Engine Pipeline Architecture:

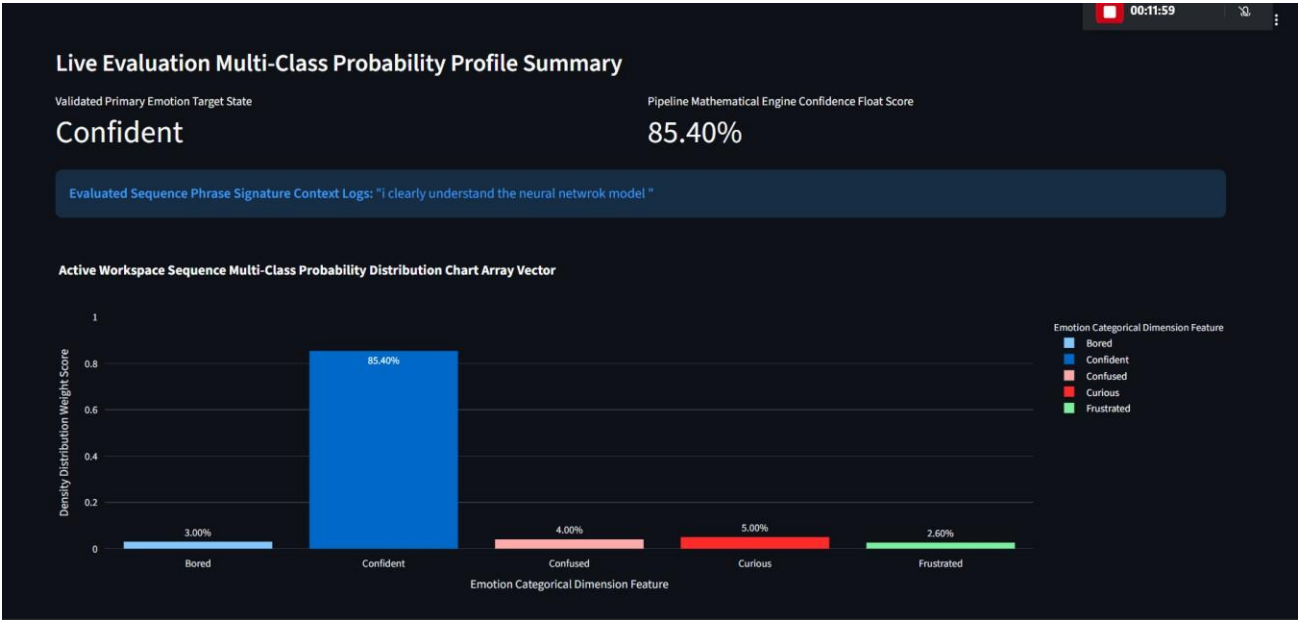
☒ BiLSTM ☐ BERT

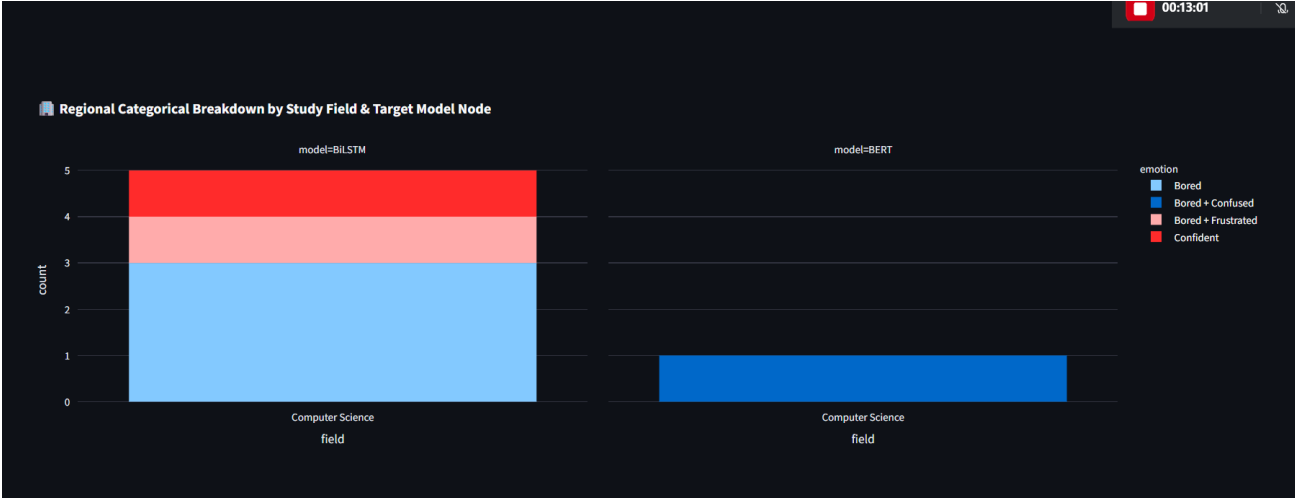
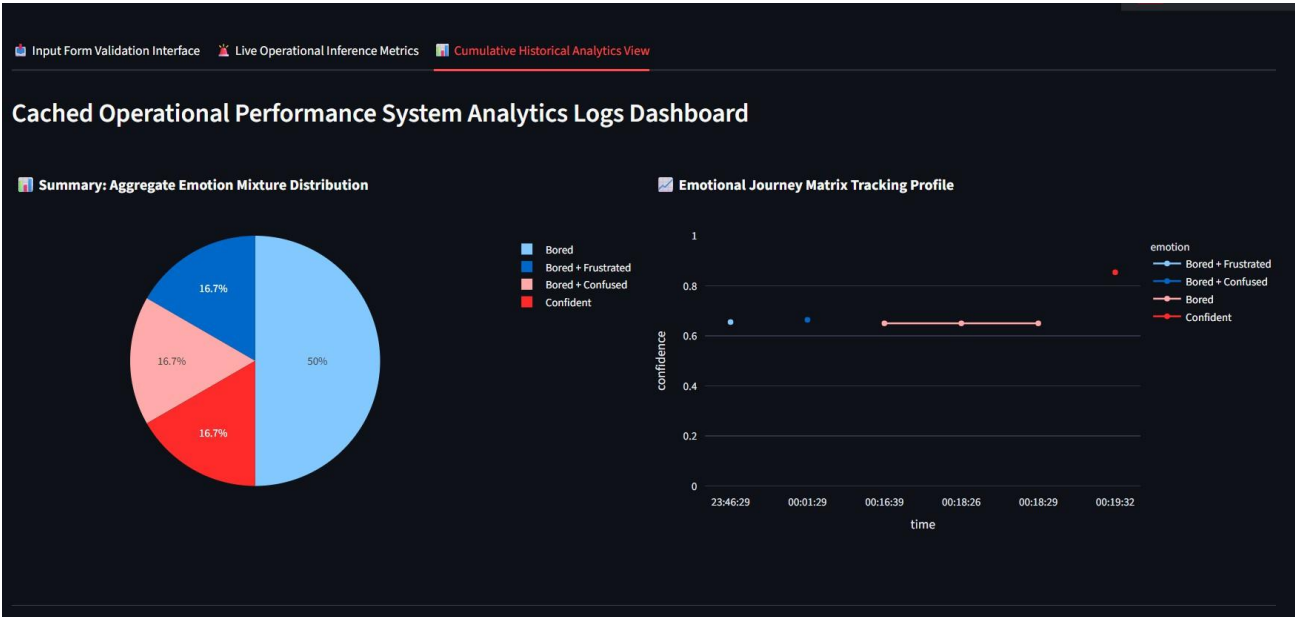
Describe your active academic task problem challenge profile query details here:

i clearly understand the neural network model

Get Learning Support Pipeline Prediction

Verification cycle computed! Check live metric profiles in next panels.





📄 System Session Activity Query Audit Log Logs

	emotion	timestamp	confidence	field	model
0	Bored + Frustrated	2026-06-30 23:46:29	0.655	Computer Science	BilSTM
1	Bored + Confused	2026-07-01 00:01:29	0.664	Computer Science	BERT
2	Bored	2026-07-01 00:16:39	0.65	Computer Science	BilSTM
3	Bored	2026-07-01 00:18:26	0.65	Computer Science	BilSTM
4	Bored	2026-07-01 00:18:29	0.65	Computer Science	BilSTM
5	Confident	2026-07-01 00:19:32	0.854	Computer Science	BilSTM

https://drive.google.com/drive/folders/1JJmYiLRtv6E1YXw2dM87fnRPSmwxakj2?usp=drive_link

Click here to view the system demonstration